RABBITS AND WILDCATS IN THE GREATER CÔA VALLEY

An open connectivity screen for trophic rewilding in northern Portugal

Linda Angulo Lopez · Rewilding Portugal · ORCID 0009-0009-5689-7148

THE QUESTION

Can the rabbit, the wildcat's main prey, be restored inside the corridors the wildcat already uses?

The European rabbit (Oryctolagus cuniculus) is the keystone prey of the Iberian Peninsula, and Endangered in its native range (IUCN, 2019); the European wildcat (Felis silvestris) depends on it. Where apex predators return, they can hold foxes and other mid-sized predators in check (Jiménez et al., 2019); where rabbits crash, those predators switch to other prey (Finlayson et al., 2021). The Côa has no lynx: can the wildcat, with its prey restored, play part of that role?

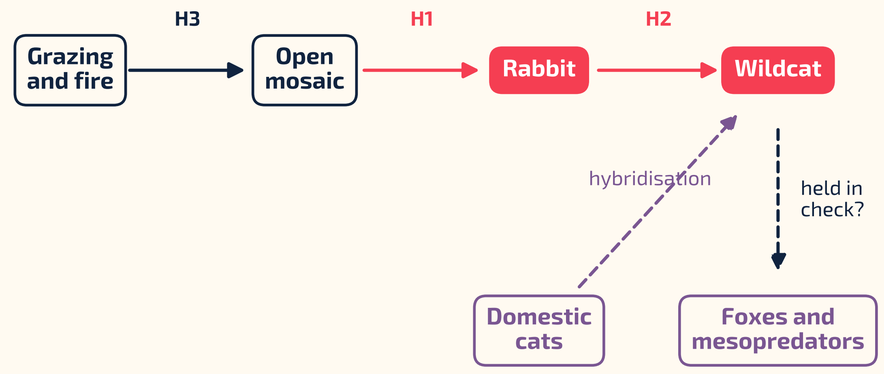
H1 Rabbits follow the grazing mosaic.

H2 Wildcat habitat tracks rabbit habitat.

H3 Fire resets the rabbit mosaic, for better and worse.

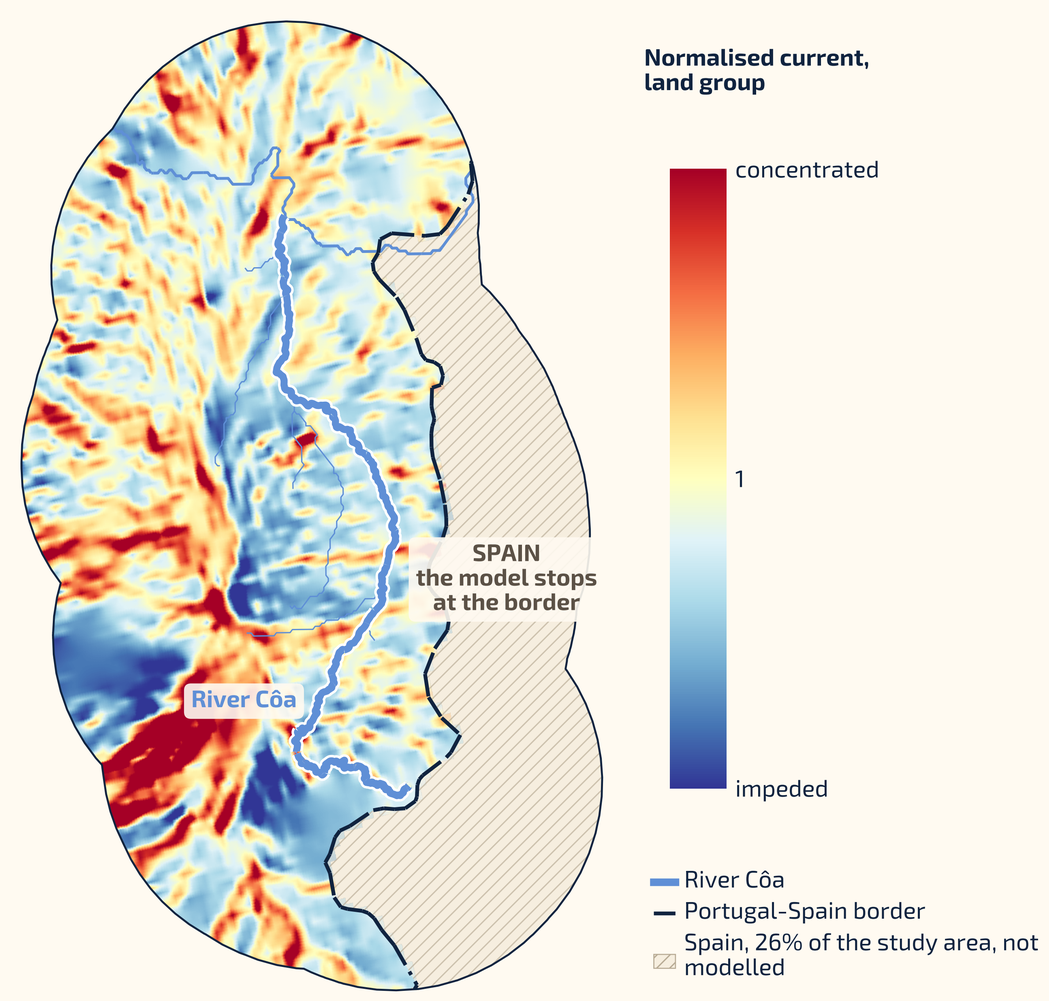
Hypotheses from the synthesis preprint (Angulo Lopez, 2026a, §6), tested with the connectivity workflow (Angulo Lopez, 2026b).

TWO SPECIES, ONE CHAIN



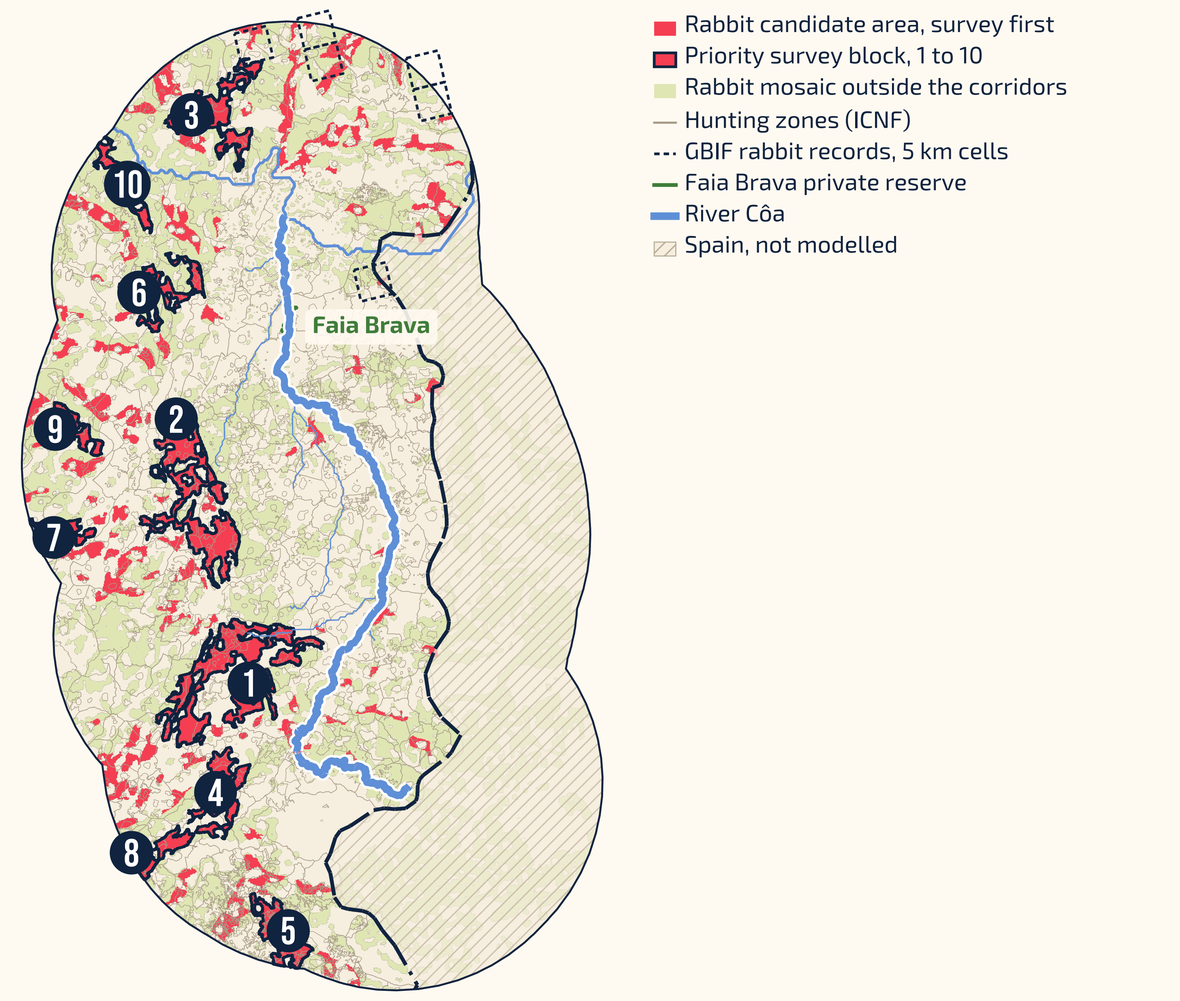
1 WHERE THE WILDCAT CAN MOVE

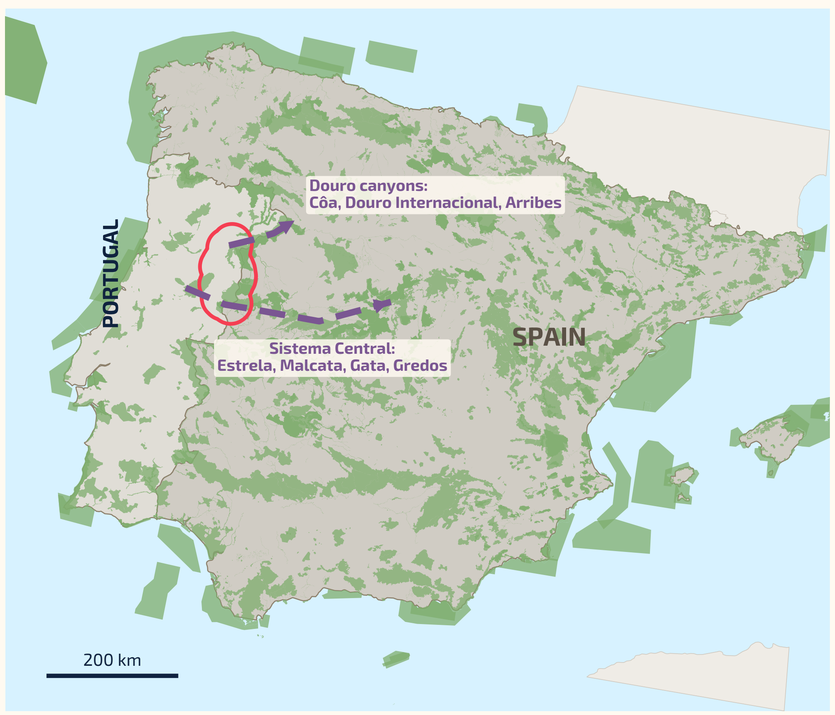
Land connectivity for wolf, wildcat and red deer, fire-aware: warm colours show where movement concentrates. Redrawn from the same raster as Fig. 4 of the workflow preprint (Angulo Lopez, 2026b). GBIF holds no wildcat record in the study area, so the corridor is modelled, not observed.



2 WHERE TO RESTORE RABBITS

Rabbit mosaic inside the top third of those corridors: 1,045 km² in blocks of 1 km² or more. The ten priority survey blocks carry the most corridor current, ranked 1 to 10. Screening maps, not release sites: every block needs a field survey.





WHY IT MATTERS FOR IBERIAN CONNECTIVITY AND REWILDING

(i) **A crossroads of two transboundary axes.** The Sistema Central (Estrela, Malcata, Gata, Gredos) runs east into Spain; the Douro canyons (Côa, Douro Internacional, Arribes del Duero, in the Meseta Ibérica Biosphere Reserve, UNESCO 2015) run north. Proposed axes, not modelled: the model stops at the border.

(ii) **The links lie between protected sites.** Natura 2000 covers 26% of the modelled study area but only 12% of the wildcat's top corridors and 9% of the rabbit candidate area. Restoration on hunting land between sites is what joins the network (Habitats Directive, Art. 10).

(iii) **Stepping stones at the western end of the Sistema Central.** 4 of the five blocks with most corridor current lie within 4 km of Serra da Estrela or Malcata.

(iv) **Prey first, predators follow.** The rabbit feeds the wildcat (Annex IV, Habitats Directive) and Annex I raptors such as Bonelli's eagle and the eagle-owl, and it is the prey base of the Iberian lynx, reintroduced in Portugal's Guadiana valley since 2015.

(v) **Restoration law.** National restoration plans under the Nature Restoration Regulation (EU) 2024/1991 are a place for these cross-border links.

EEA Natura 2000, end-2024, SPA and SCI; Natural Earth.

1,045 km² of rabbit candidate area inside the corridors

10 priority survey blocks

87% of the candidate area on hunting land

20 GBIF rabbit records, in 6 cells of 5 km

0 GBIF wildcat records in the study area

26% of the study area in Spain, not yet modelled

METHOD

(i) **Study area.** 10,230 km² on a 100 m grid: the Côa and 30 km around it, extended to its catchment.

(ii) **Ten species, three groups** (land, water, air), after Prima et al. (2024). The wildcat is in the land group, with the wolf and red deer. Suitability from GBIF records (Random Forest).

(iii) **Circuit theory in pure Python,** with penalties for fire (MODIS, 2015 to July 2025), 22 barriers recorded in the field and two UNESCO areas (Angulo Lopez, 2026b).

(iv) **Rabbit screen.** At least 20% grassland or cropland and 20% tree cover or shrubland within about 500 m (ESA WorldCover 2021), under 5% built, not burned since 2023, 200 m or more from a road, in the top third of the land current.

(v) **Limits.** Training accuracy of 0.78 to 0.99 is not validation.

WHAT THE MAPS SHOW

(i) **Mosaic is common, corridors are not.** Rabbit mosaic covers 4,410 km²; 1,045 km² of it lies inside the wildcat's corridors.

(ii) **Two blocks carry a quarter.** The ten blocks range from 23 to 145 km²; the two largest hold 25% of the candidate area.

(iii) **Hunters hold the land.** 87% of the candidate area is in hunting zones. Faia Brava is 88% rabbit mosaic but carries little corridor current, so none of it is a candidate.

(iv) **Recent fire removes little:** 0.2% of the study area burned from 2023 on, so H3 cannot yet be tested.

(v) **Habitat, not counts.** 20 rabbit records and no wildcat record in GBIF: the screen needs field data.

PROTECTED, NOT RELEASED

No wildcats are released. The existing population is protected, and the work goes into its prey and its neighbours. Wildcat records are generalised to 5 km before they reach a map.

(i) **Keep it wild.** Hybridisation with domestic cats is the main threat, with persecution (Angulo Lopez, 2026a): neuter domestic cats near villages.

(ii) **Restore its prey,** inside its corridors (map 2).

(iii) **Watch the mesopredators.** Where rabbits crash, foxes and cats switch to other prey (Derham et al., 2018; Finlayson et al., 2021). Monitoring starts before any rabbit release.

WHAT WE PROPOSE

(i) **A rabbit baseline** in the ten priority blocks, with the hunters (H1, H3).

(ii) **Cat neutering and mesopredator monitoring** before and during any rabbit release.

(iii) **Camera traps on a 5 km grid** to test H2.

(iv) **The rabbit as a focal species** in the workflow, then rerun the corridors.

(v) **Extend the model into Spain:** Campo de Azaba, El Rebollar, Arribes del Duero.

WHY THE CÔA

The river is free-flowing because a 1995 campaign, "As gravuras não sabem nadar", stopped a dam to save its rock art (suspended 17 January 1996; UNESCO, 1998).

Which predators will the next chapter of the Côa include, and who decides?

Angulo Lopez, L. (2026a). Synthesis and research agenda for trophic rewilding in the Greater Côa Valley (hypotheses H1–H3). Preprint, Zenodo, doi:10.5281/zenodo.22997241. Angulo Lopez, L. (2026b). Movement, Water, Connection: a reproducible Python connectivity workflow and field illustration for the Greater Côa Valley rewilding landscape, Portugal. Preprint, Zenodo, doi:10.5281/zenodo.22979020. Data: Côa Valley Eco-Connectivity v6, Etalab 2.0, data.gouv.fr/datasets/coa-valley-eco-connectivity-v6. Poster code: github.com/coa-connectivity-lab/ser2027. Derham, T. et al. (2018). Phil. Trans. R. Soc. B 373, doi:10.1098/rstb.2018.0127. Finlayson, G. et al. (2021). Restoration Ecology 30, doi:10.1111/rec.13552. Jiménez, J. et al. (2019). Biological Conservation, doi:10.1016/j.biocon.2019.108234. Prima, M.-C. et al. (2024). Methods in Ecology and Evolution. Villafuerte, R. and Delibes-Mateos, M. (2019). Oryctolagus cuniculus. IUCN Red List, e.T41291A45189779. ESA WorldCover 10 m 2021 v200, CC BY 4.0. HydroRIVERS v1.0. NASA MODIS MCD64A1. OpenStreetMap contributors (ODbL). ICNF hunting zones. DGPC heritage boundaries. Natural Earth. GBIF.org occurrences, generalised to 5 km. Resolução do Conselho de Ministros n.º 14/1996. Maps: EPSG:3763; every figure is computed in scripts/poster.py from the v6 bundle. Code written with AI assistance (Claude, Anthropic) from the author's method and direction, and reviewed by the author. Reviewer: VRConservation (GitHub). Linda Angulo Lopez, 2026, poster v0.2. Text and maps CC BY 4.0; code MIT.